

Module 4

The Time Value of Money

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Goals

- **Part 1**
 - Translate dollars today into dollars tomorrow (“future value”)
 - Translate dollars tomorrow into dollars today (“present value”)

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- **Part 2**
 - Financial calculators

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- **Part 3**

- Special cases: perpetuities, annuities, and their variants

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- **Part 1**
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 - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
 - Financial calculators
- **Part 3**
 - Special cases: perpetuities, annuities, and their variants
- **Part 4**
 - The role of inflation

Module 4, Part 1

Future Value and Present Value

The Time Value of Money



Would you rather receive a dollar today, or a dollar tomorrow?

The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



The Time Value of Money

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- Why?
- Because we can invest that dollar today and collect interest on our investment!
 - Assuming, there is some positive rate of return investment for us to pursue

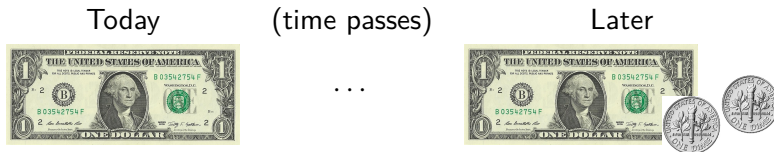


Future Values

- **Future value**, the amount to which an investment will grow after earning interest
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$$\text{Future value (FV)} = \text{PV} \times (1+r)^T$$

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$$\text{Future value (FV)} = \text{PV} \times (1 + r)^T$$

$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

Compound Interest

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- Compound interest is the default in finance (e.g., deposit rate)

Future Value, Compound Interest

- You have \$100 in a bank account, and your bank pays 6% per year in interest
- How much interest will you have earned after five years?

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Year	Start balance	Compound interest	End balance
1	\$100.00	$0.06 \times 100.00 = \$6.00$	\$106.00
2	\$106.00	$0.06 \times 106.00 = \$6.36$	\$112.36
3	\$112.36	$0.06 \times 112.36 = \$6.74$	\$119.10
4	\$119.10	$0.06 \times 119.10 = \$7.15$	\$126.25
5	\$126.25	$0.06 \times 126.25 = \$7.57$	\$133.82

Future Value, Compound Interest

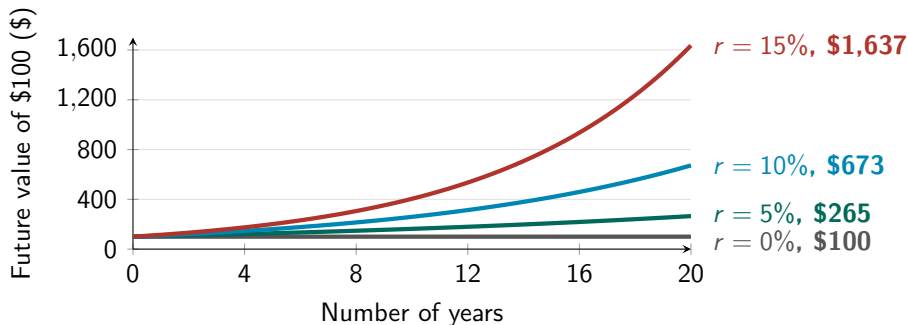
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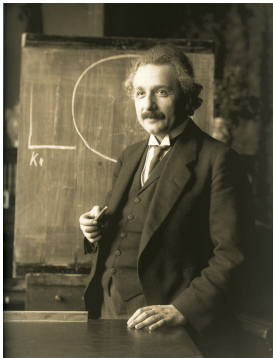
- With compound interest, your \$100 grows to $\$100(1 + r)^T = 133.82$, so you earned \$33.82 in compound interest

Future Value

- The future value (FV) depends heavily on the interest rate r , especially as the time dimension T grows

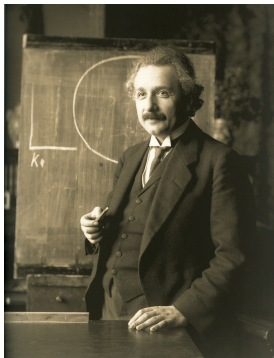


What's the Most Powerful Force in the Universe?



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- Einstein, ...
- “Compound interest”

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$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

Holding Period Return

- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for T periods with returns $r_1, r_2, \dots, r_T$ gives $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \dots (1 + r_T)$

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$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 1.80 \times 0.80 - 1 = 44\%$$

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- Notice that the annual return is *not* $\frac{80\% - 20\%}{2} = 30\%$

Recap, Three Kinds of Returns

Return	Holding period return	Annualized return
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

Recap, Three Kinds of Returns

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Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

- These financial returns are *not directly related in any way* with **Module 3's** accounting-based ROA and ROE measures

Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?



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b. $FV = \$24 \times (1 + 0.035)^{395} = \19 million

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- 11.896 years, because $(1 + 0.06)^{11.896} = 2$
 - We will learn how to calculate this with the financial calculator



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$$\text{Present value (PV)} = FV \times \frac{1}{(1 + r)^T}$$

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- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value

Present Value

- **Discount rate**, the rate used to compute PVs (i.e., r)
 - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
 - For a risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**

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- **Discount factor**, the present value of \$1 received in year T when the discount rate is r , less than one when r is positive

$$\text{Present value (PV)} = \text{FV} \times \frac{1}{(1+r)^T}$$

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 **Discount factor**

- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

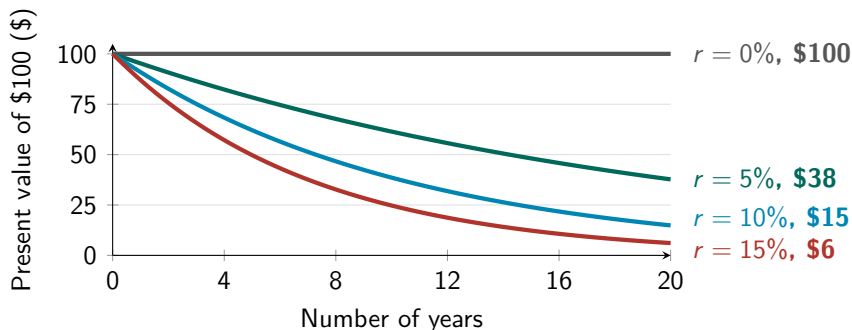
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

Present Value

- The present value (PV) of \$100 in the future depends heavily on the discount rate r , especially as the time dimension T grows



Present Value, Multiple Future Cash Flows

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$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

- where C_1 is the cash flow in one year, C_2 is the cash flow in two years, etc.

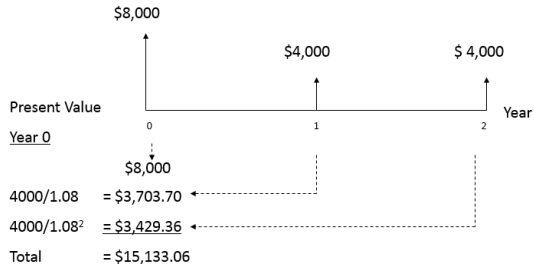
Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
 - A. Pay \$15,500 cash now, or
 - B. Make three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

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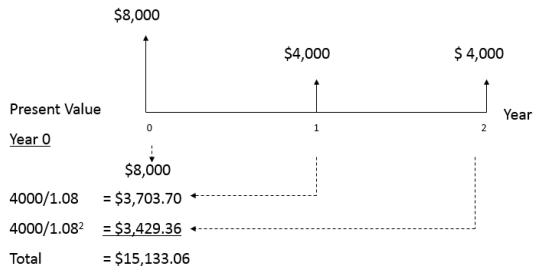
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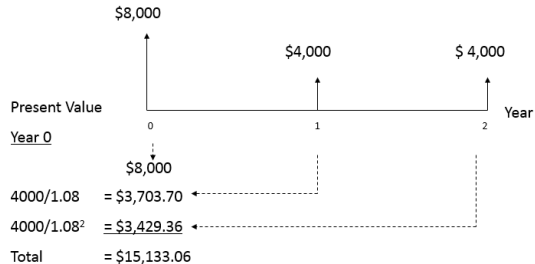
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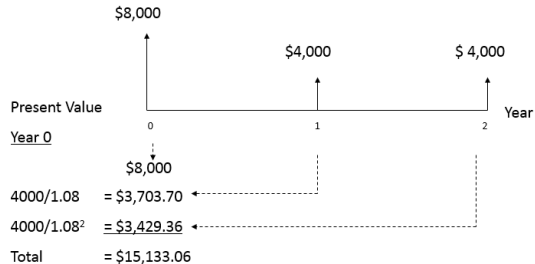


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- The present value (\$15,133.06) is less than \$15,500, so choose the payment plan
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

Recap of Part 1

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- The present value of a future cash flow is the value of that cash flow multiplied by its discount factor

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- A dollar today is worth more than a dollar tomorrow
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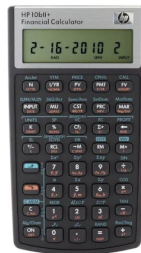
- Present values can be added together to calculate the total present value of multiple future cash flows

Module 4, Part 2

Financial Calculators

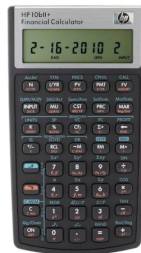
Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)



Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)
- A complement to the underlying math and intuition, never a substitute



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- After you reset, you should repeat the next two settings!

Set the Number of Significant Digits to at Least 4!

- On the HP 10bII+, to change the number of digits displayed to 4 decimals, press
 - OrangeOption
 - = (DISP)
 - 4

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 - .
 - 4
 - ENTER

Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1,
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- To change back to the factory default (12), press
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 - OrangeOption
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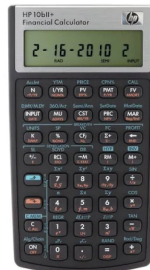
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- You must do this!

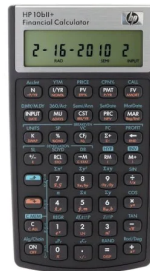
Financial Calculators, Basic Functionality

- N, number of periods (usually years)



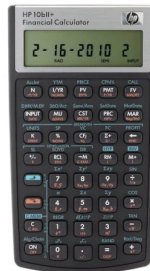
Financial Calculators, Basic Functionality

- N , number of periods (usually years)
- I/YR , interest rate, expressed as a percentage (not a decimal)



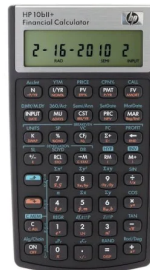
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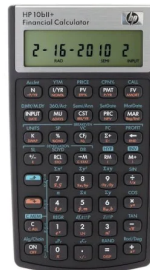
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n	i	PV	PMT	FV
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- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
- It did so by selling bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each (they sold around $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$ bonds, and thus promised to pay back around \$27.5 billion in total)
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- Bonus, what would the payment be if the monthly interest was 0.2%?

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- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?

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- Let's try to solve some of the previous examples using Excel!

Example, A-Rod's \$275 Million Contract with the Yankees

Year	Salary	"Signing bonus"
2008	27	2
2009	32	1
2010	32	1
2011	31	1
2012	29	1
2013	28	1
2014	25	3
2015	21	
2016	20	
2017	20	
Total	265	10

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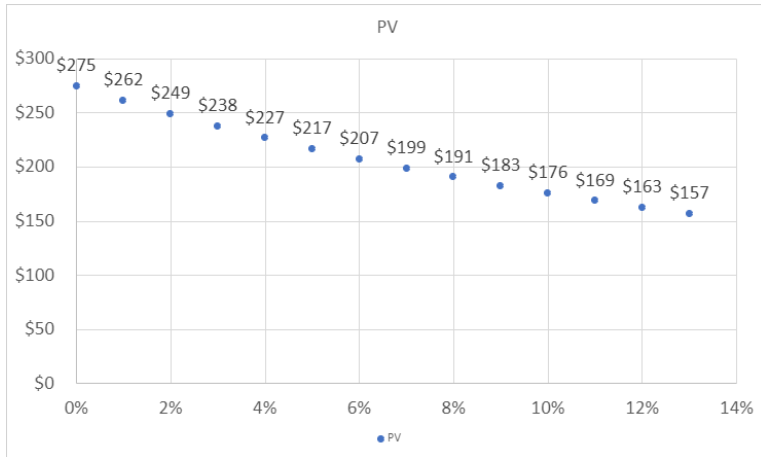
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- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?
- Why might it be useful for A-Rod to know the PV, and not simply the (undiscounted) sum of all future payments?



The PV of Payments to A-Rod



Recap of Part 2

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- The PV has the opposite sign of the payments (PMT) and the future value (FV)
- Set the number of compounding periods per year to 1, and show at least 4 significant digits
- Excel has the same functionality, with the rate in decimals

Module 4, Part 3

Special Cases: Perpetuities and Annuities

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- Why care about perpetuities and annuities?
 - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)
 - There exist neat shortcuts that make it very easy to calculate the present value of these patterns of cash flows

Perpetuity Structure

- A perpetuity is a stream of equal cash flows (C) that occur at regular intervals and last forever



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- C is the annual cash flow (aka the payment), and r is the discount rate

Example, Perpetuities

- Recall A-Rod's Yankees contract. Suppose the discount rate is 10%, then the PV of the contract (paying \$275m over 10 years) is \$175.6m
- Now suppose the Yankees offer A-Rod two options,
 - A. The contract as previously described (10 years with \$275m in total), or
 - B. A perpetual payment of C dollars every year. He will play for 10 years, but continue to be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- How much should the Yankees offer as the perpetual payment C for him to be **indifferent** between contract (A) and the “being paid forever” contract (B)?



Example, Perpetuities

- Solution, he is indifferent if the contracts have the same present value!

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$$PV_{\text{Contract A}} = PV_{\text{Contract B}} = \frac{C_B}{r} \Rightarrow C_B = PV \times r = 175.6 \times 10\% = \$17.56m$$

Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



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Perpetuity, Timing

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- **Ordinary perpetuity**, first payment at time 1

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- **Delayed perpetuity**, first payment at time $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at C but grows at a constant growth rate g forever. This is a growing perpetuity

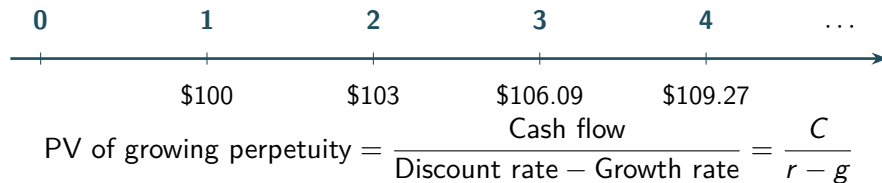
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- Example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has cash flows



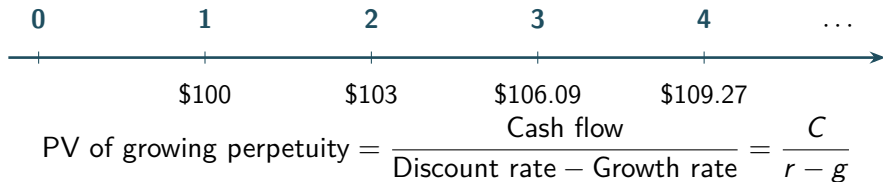
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- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Modules 5 and 9](#))
- When $g = 0$, the PV simplifies to the formula for a regular (non-growing) perpetuity

Annuities

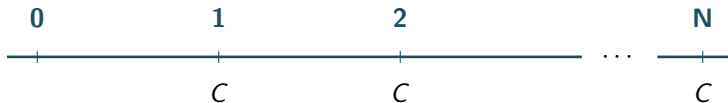
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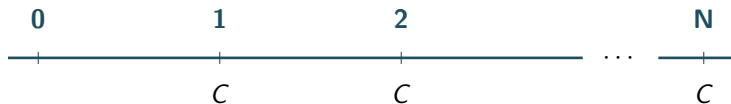


- Note, the first payment happens after one period (not immediately)
- Present value of an annuity,

$$\text{PV of annuity} = C \underbrace{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)}_{\text{present value annuity factor (PVAF)}}$$

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- C is the annual cash flow (aka the payment), r is the discount rate, and N is the number of years that the annuity lasts

Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

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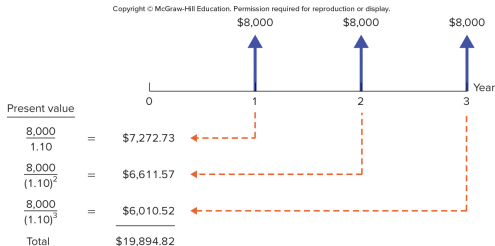
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$$PV = C \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left(1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

Example, Annuity

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Example, Annuity

- Suppose the discount rate for A-Rod's contract is 10%, so the present value is \$175.6m
- Now suppose instead the Yankees offer an annuity for 10 years with the same present value
- How much would he get paid per year, and in total (undiscounted) over 10 years, under such a same-value annuity contract?



Example, Annuity

- Solution,

Example, Annuity

- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

Example, Annuity

- Solution,

$$175.6 = C \frac{1}{10\%} \left(1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

- So he would be paid \$28.58m per year, or \$285.8m in undiscounted total over 10 years (slightly more in total, because his actual contract is front-loaded, which raises its present value compared to a constant payment)

Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

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Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



Perpetuities and Annuities, Timing

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- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



- **Ordinary annuity**, first payment at time 1

$$PV = PV \text{ annuity}$$



- **Delayed annuity**, first payment at time $t > 1$

$$PV = PV \text{ annuity} \times \frac{1}{(1+r)^{t-1}}$$



Practice Problem

- A store offers two payment plans. Under the installment plan, you pay 25% down and 25% of the purchase price in each of the next 3 years. If you pay the entire bill immediately, you can get a discount of 7% on the purchase price. Assume the product sells for \$100
- a-1. Calculate the present value of the payments if you can borrow or lend funds at an interest rate of 4%
- a-2. Which is a better deal?
- b-1. Calculate the present value if the payments on the 4-year installment plan do not start for a full year
- b-2. Which is a better deal?

Practice Problem

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$$PV = \$25 + [\$25 \times \text{annuity factor (4\%, 3 years)}]$$

$$PV = \$25 + \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^3} \right] = \$94.38$$

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- a-2. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator, $PMT = -25$, $N = 3$, $FV = 0$, $I/YR = 4$, leading to 69.38, and next, $69.38 + 25 = 94.38$

Practice Problem

- b-1. Now it is an ordinary annuity, since the first payment is due at the end of year 1

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$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^5} \right] = \$90.75$$

Practice Problem

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$$PV = \$25 \times \left[\frac{1}{0.04} - \frac{1}{0.04 \times (1.04)^4} \right] = \$90.75$$

- b-2. The installment plan is better (\$90.75 beats \$93)
- On a financial calculator, $PMT = -25$, $N = 4$, $FV = 0$, $I/YR = 4$

Practice Problem

- A local bank advertises the following deal, “Pay us \$100 at the end of each year for 11 years, and then we will pay you (or your beneficiaries) \$100 at the end of each year forever”
 - a. Calculate the present value of your payments to the bank if the interest rate is 7.00%
 - b. What is the present value of a \$100 perpetuity deferred for 11 years if the interest rate is 7.00%?
 - c. Is this a good deal?

Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

Practice Problem

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$$PV = C \left(\frac{1}{r} \right) \left(1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left(\frac{1}{7\%} \right) \left(1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

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- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV

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- b. The bank's payments to you are a perpetuity deferred for 11 years

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- On a calculator, PMT = 100, I/YR = 7, N = 11, FV = 0. Find PV
- b. The bank's payments to you are a perpetuity deferred for 11 years

$$PV_{11} = \frac{C}{r}$$
$$PV_{11} = \frac{100}{7\%} = \$1,428.57$$
$$PV = \frac{PV_{11}}{(1+r)^{11}} = \frac{1,428.57}{(1+7\%)^{11}} = \$678.70$$

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- c. Not a good deal (you pay \$749.83 for something worth \$678.70)

Growing Annuities

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$$\text{PV of growing annuity} = \frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right]$$

Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
 - A. A lump sum of \$327 million, or
 - B. 30 installments of a growing annuity due with $g = 4\%$, starting at \$9.43m, with 30 payments, totaling around \$529 million



Photo, AP/Mark Humphrey

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- Which option should they take?



Photo, AP/Mark Humphrey

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ($g = 4\%$), it starts immediately and grows for another 29 years at 4% and then ends
- Suppose the discount rate is 3%

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
28	28.28
29	29.41
Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
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Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$

Year	Payout
0	9.43
1	9.81
2	10.20
...	...
27	27.19
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Total	528.88

Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

$$\begin{aligned} & \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^N \right] \times (1+r) \\ &= \frac{9.43}{0.03-0.04} \left[1 - \left(\frac{1+0.04}{1+0.03} \right)^{30} \right] \times (1+0.03) \\ &= \$326.58\text{m} \end{aligned}$$

Year	Payout
0	9.43
1	9.81
2	10.20
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27	27.19
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Take the Lump Sum or Growing Annuity?

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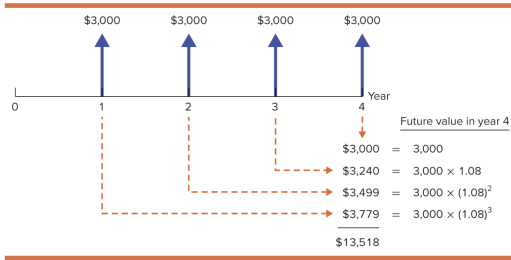
$$\frac{C}{r - g} \left[1 - \left(\frac{1 + g}{1 + r} \right)^N \right] \times (1 + r)$$
$$= \frac{9.43}{0.03 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$
$$= \$326.58\text{m}$$

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2	10.20
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27	27.19
28	28.28
29	29.41
Total	528.88

- That is, the present value of the annuity is almost exactly the same as (but slightly lower than) the \$327m lump sum

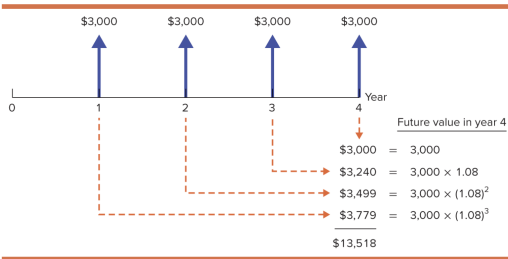
Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



Future Value of Annuities

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$$\text{FV of annuity of \$1 a year} = \text{PV of annuity of \$1 a year} \times (1 + r)^N$$

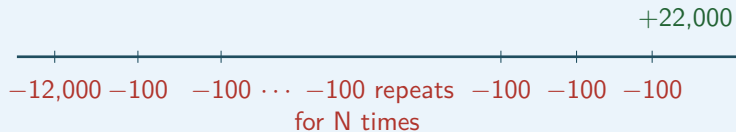
$$= \left[\frac{1}{r} - \frac{1}{r(1 + r)^N} \right] \times (1 + r)^N = \frac{(1 + r)^N - 1}{r}$$

Practice Problem

- I now have \$12,000 in the bank earning interest of 0.50% per month. I need \$22,000 to make a down payment on a house. I can save an additional \$100 per month. How long will it take me to accumulate the \$22,000?

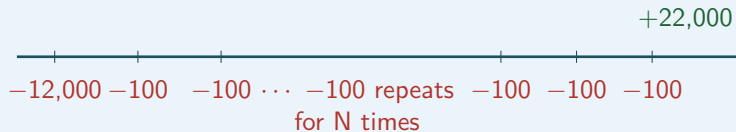
Practice Problem

- From your perspective, the cash flow is



Practice Problem

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- You need to solve, future value (FV) of the \$12,000 + FV of the N \$100s = FV of the \$22,000

$$12000 \cdot (1 + 0.005)^N + 100 \cdot \frac{(1 + 0.005)^N - 1}{0.005} = 22000$$

Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

Practice Problem

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	N	I/Y	PV	PMT	FV
Solve for	54.52				

- Let's recall that financial calculators are programmed to solve the following,

$$PV + \left[\frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

Practice Problem

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Enter		0.50	-12,000	-100	22,000
	N	I/Y	PV	PMT	FV
Solve for	54.52				

Practice Problem

- A couple will retire in 50 years. They plan to spend about \$20,000 a year in retirement, which should last about 25 years. They believe that they can earn a real interest rate of 9% on retirement savings
 - a. If they make annual payments into a savings plan, how much will they need to save each year? Assume the first payment comes in 1 year
 - b. How would the answer to part (a) change if the couple also realize that in 20 years they will need to spend \$50,000 on their child's college education?

Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



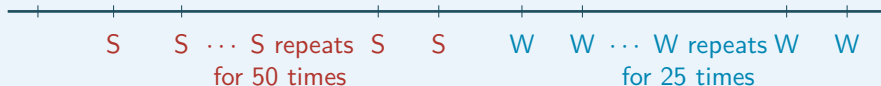
Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



- How much savings is required at the time of retirement?

$$PV_{\text{withdrawals, year 50}} = 20,000 \frac{1}{0.09} \left(1 - \frac{1}{(1 + 0.09)^{25}} \right) = 196,451.59$$

- On a calculator, $PMT = 20,000$, $I/YR = 9$, $N = 25$, $FV = 0$

Practice Problem

- The FV of the 50 savings, each of the amount $\$S$, should be that

Practice Problem

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$$FV_{\text{savings, year 50}} = S \left(\frac{(1 + 0.09)^{50} - 1}{0.09} \right) = 196,451.59 \Rightarrow S = \$241.02$$

- On a calculator, $FV = 196,451.59$, $I/YR = 9$, $N = 50$, $PV = 0$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, $PV \text{ of all savings, discounted to today} = PV \text{ of all withdrawals, discounted to today}$

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- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, PV of all savings, discounted to today = PV of all withdrawals, discounted to today
- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

- Now determine the annual savings for 50 years that equates to that present value

$$\$11,563.53 = C \times \frac{1 - \frac{1}{(1+9\%)^{50}}}{9\%} \Rightarrow C = \$1,054.90$$

- On a calculator, PV = 11,563.53, I/YR = 9, N = 50, FV = 0

Practice Problem

- Professor's Annuity Corporation offers a lifetime annuity to retiring professors. For a payment of \$79,000 at age 65, the firm will pay the retiring professor \$575 a month until his death
- a. If the professor's remaining life expectancy is 20 years, what is the monthly interest rate on this annuity?
- b. If the monthly interest rate is 1.00%, what monthly annuity payment can the firm offer to the retiring professor?

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

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$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $PMT = 575$, $N = 240$, $PV = -79,000$, $FV = 0$. Find I/YR

Practice Problem

- a. The monthly payments are a 240-month ordinary annuity

$$\$575 \times \left[\frac{1}{r} - \frac{1}{r \times (1+r)^{240}} \right] = \$79,000 \Rightarrow r = 0.52\%$$

- On a calculator, $\text{PMT} = 575$, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$. Find I/YR
- b.

$$C \times \left[\frac{1}{0.01} - \frac{1}{0.01 \times (1.01)^{240}} \right] = \$79,000 \Rightarrow C = \text{PMT} = \$869.86$$

- On a calculator, $N = 240$, $\text{PV} = -79,000$, $\text{FV} = 0$, $\text{I/YR} = 1$. Find PMT

Recap of Part 3

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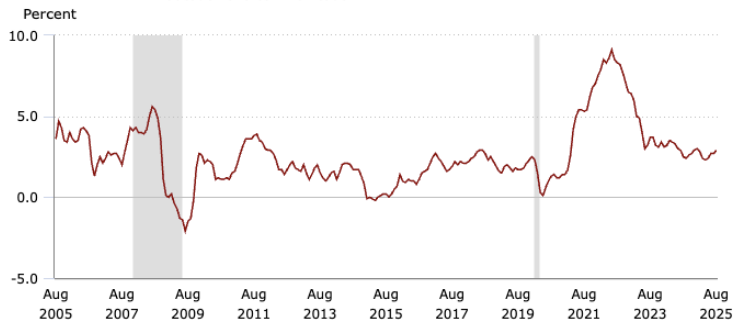
Recap of Part 3

- A perpetuity is a level stream of cash flows at regular intervals that never ends
- An annuity is a level stream of cash flows at regular intervals that ends after N periods (usually years, but the periods do not have to be years)
- These have handy formulas for calculating their present values!
- Timing,
 - Ordinary perpetuities and annuities make their first payment at time $t = 1$
 - Perpetuity and annuity dues make their first payment at $t = 0$
 - Delayed perpetuities and annuities make their first payment at $t > 1$

Module 4, Part 4

Inflation

History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
Source: U.S. Bureau of Labor Statistics.

Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)

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 - Compare dollars at time $t = 0$ to dollars at time $t = 1$

Inflation

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- **Nominal interest or nominal return**, the rate at which money invested grows
 - Compare dollars at time $t = 0$ to dollars at time $t = 1$
- **Real interest or real return**, the rate at which purchasing power increases
 - Compare the ability to buy a basket of goods at time $t = 0$ to the ability to buy the same basket of goods at time $t = 1$

Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

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Approximation, $\text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$

- Inflation also affects discounting,
 - We should discount nominal dollar cash flows using a nominal discount rate
 - We should discount real dollar cash flows using a real discount rate

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?

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- Answer,

$$1 + \text{Real interest rate} = \frac{1 + \text{Nominal interest}}{1 + \text{Inflation}} = \frac{1.06}{1.02} = 1.039$$

Example, Inflation

- Question, if the interest rate on one-year government bonds is 6.0% and the inflation rate is 2.0%, what is the real interest rate?
- Answer,

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- Approximation, $\text{real} \approx 6\% - 2\% = 4.0\%$

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- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

Thank you!

Please let me know if you have any questions.